TATA Tele Business Services | DO Big

AI BASED CHATBOT

ARTIFACTS

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Submitted By

Tata Tele Business Service

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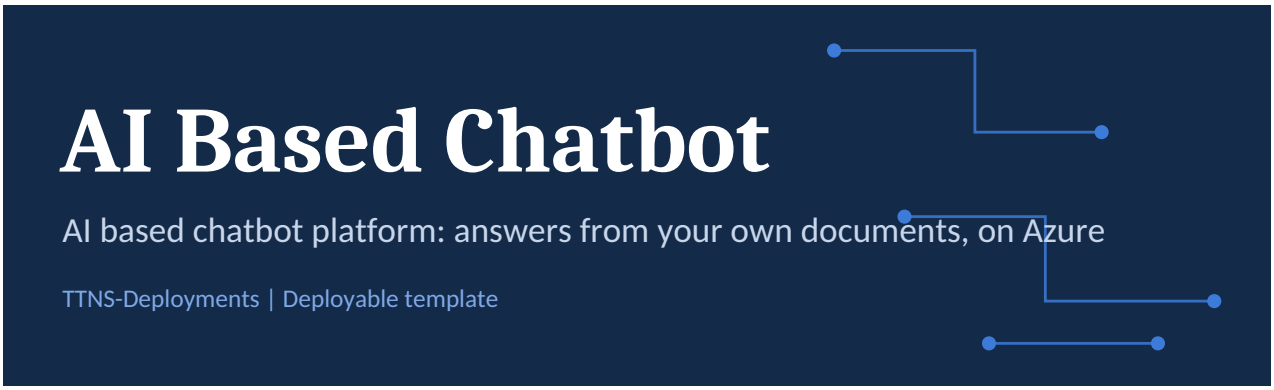
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1. Document objective

- The README for a project called **AI Based Chatbot**, which sets up a chatbot on Azure that answers questions from an organisation's own documents.
- **How it's built:** One Azure template creates the whole environment. You start it with the **Deploy to Azure** button in the Azure portal, or with one script command (Bash or PowerShell), so nobody has to set up each service by hand.
- **The application:**
  - A Python web app on Azure App Service shows the chat page and handles the questions.
  - Azure AI Search finds the parts of your documents that match each question, by exact words and by meaning.
  - Azure OpenAI writes the answer from those parts only and lists the sources it used.
  - A storage account holds the documents. Azure AI Search reads them and keeps its index up to date automatically.
- **Supporting services:**
  - Key Vault for the one secret (the key that protects document upload)
  - Application Insights and Log Analytics for logging and monitoring
  - Managed identities, so the services connect to each other without passwords or keys
- **Intended use:** A ready-to-deploy starting point. You choose the size of each service to fit your budget and traffic. It has been tested on a developer machine but not yet on a live Azure subscription, so deploy it to a test subscription first.
- **What the document covers:**
  - An overview of the solution and how it works
  - What you need before you start
  - How to choose the size of each service
  - The ways to deploy it: the button, Bash, PowerShell and GitHub Actions
  - What to do afterwards: check that it works and add your documents
  - How to use the chatbot
  - Cost, security and clean-up notes
  - Troubleshooting, the repository layout and the current status

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AI Based Chatbot



AI Based Chatbot answers questions from your organisation's own documents. People ask questions on a web page. Azure AI Search finds the most relevant passages in your documents, Azure OpenAI writes the answer from those passages, and the page shows which documents the answer came from so people can check it.

The solution is a ready-to-deploy template. You choose the size of every Azure service in one settings file, then deploy everything with one command or the **Deploy to Azure** button.

Deploy to Azure

The button creates all the Azure services and, unless you turn it off, also installs the application from the `main` branch of the GitHub repository. The deployment scripts do the same, and they also check your choices against what your Azure region offers before they start. The button only works while the GitHub repository is public.

Overview

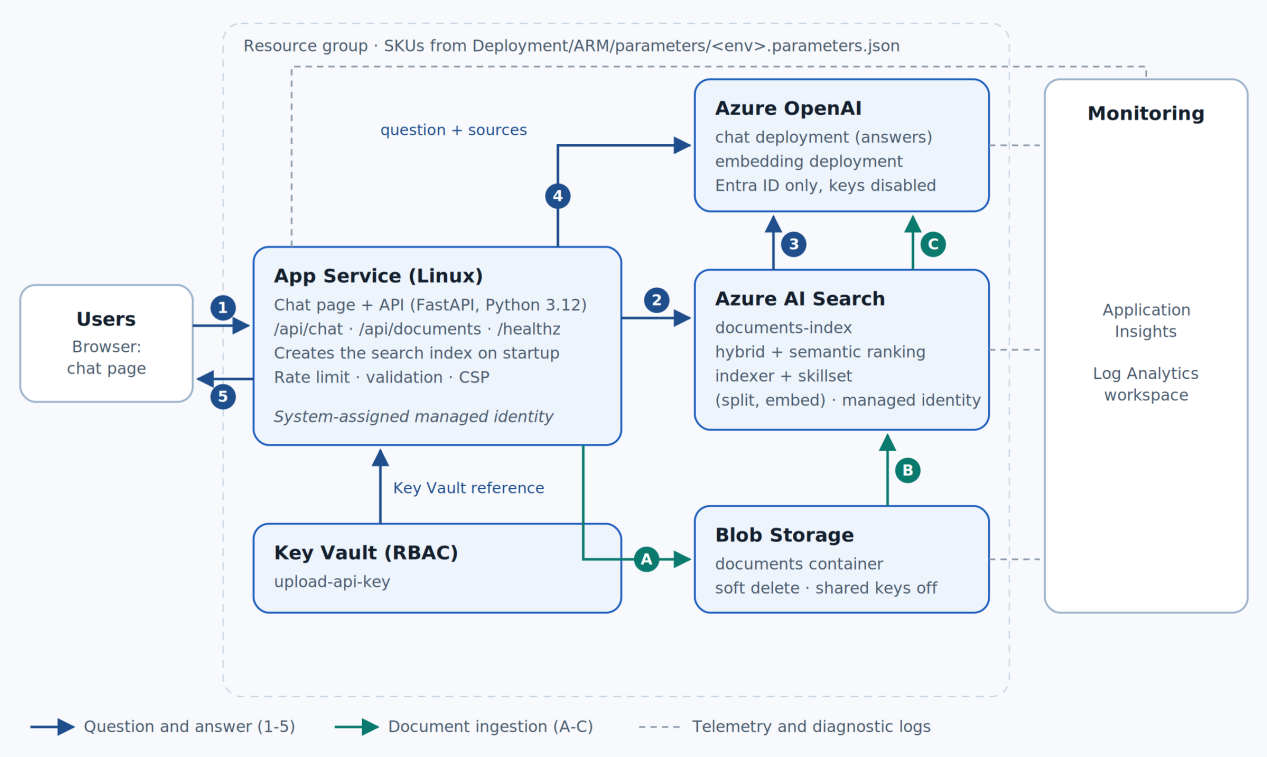
- **Answers based on your documents, with sources.** Each question is matched against your documents both by exact words and by meaning. The best passages go to the language model, which must answer only from them and mark each statement with a source number such as [1] or [2]. The page lists the sources with a short extract.
- **Follow-up questions.** The page remembers the conversation, so a follow-up such as "and for returns?" is understood.
- **Your own documents.** PDF, Word, PowerPoint, Excel, HTML, Markdown, CSV, JSON and plain text. Add them on the chat page (protected by an upload key) or with a script. Azure AI Search splits, processes and indexes them automatically.
- **No passwords or keys between services.** The services sign in to each other with managed identities, and key-based access is switched off on Azure OpenAI, Azure AI Search and Storage. The only secret, the upload key, is kept in Key Vault.
- **Sized by you.** Every service size is in a settings file with three ready-made options (`dev`, `standard` and `prod`). Your choices are checked before deployment, and the same choices appear as drop-down lists in the Azure portal.

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Terms used in this document

Term	Meaning
SKU	The size or pricing tier of an Azure service, for example B1 for a small web app plan.
Preset	A ready-made set of SKU choices (`dev`, `standard` or `prod`) stored in one file.
TPM	Tokens per minute: how much text Azure OpenAI will process for you each minute. A token is roughly three quarters of a word. Capacity is set in thousands (30 = 30,000 TPM).
Quota	The maximum TPM your Azure subscription is allowed in a region.
Replica / partition	Copies of the search service (more copies handle more questions and give an uptime guarantee) and slices of storage for the search index (more slices hold more documents).
Managed identity	An identity Azure gives a service so it can access other services without a stored password or key.
Index / indexer	The searchable copy of your documents, and the Azure AI Search process that reads your files and keeps that copy up to date.

Architecture



#	Step	Azure service
1	A user asks a question on the chat page. The browser sends it together with the recent conversation.	App Service

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#	Step	Azure service
2	The question is matched against the documents by exact words and by meaning, and the best matches are ranked.	Azure AI Search
3	Azure AI Search turns the question into numbers (an embedding) so it can be compared by meaning.	Azure OpenAI (embedding model)
4	The question and the best passages, numbered, go to the language model, which answers only from them.	Azure OpenAI (chat model)
5	The answer and the sources it used are sent back and shown on the page.	App Service
A	Someone with the upload key adds a document (or runs the ingestion script). It is stored and indexing starts.	Storage account
B	The indexer reads new and changed files. It also checks every 30 minutes.	Azure AI Search indexer
C	Each file is split into passages, and each passage is turned into an embedding and stored in the index.	Azure AI Search and Azure OpenAI

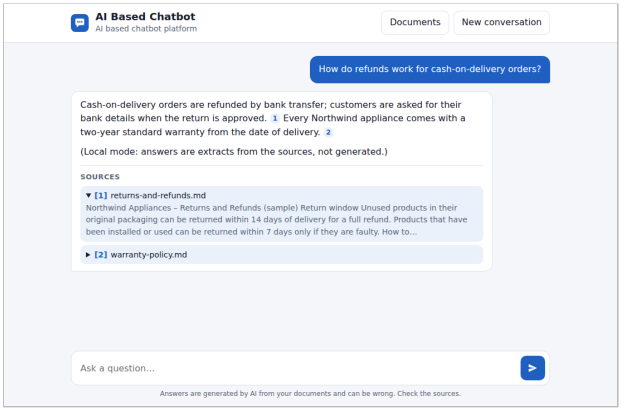
Supporting services: **Key Vault** holds the upload key. **Application Insights** and **Log Analytics** collect logs and performance data. The services connect to each other with **managed identities**, and each one gets only the access it needs.

Component	Azure resource	Settings that control its size
Web app	App Service plan (Linux) and web app, Python 3.12	`appServicePlanSku`, `appServiceInstanceCount`
Language models	Azure OpenAI with a `chat` model and an `embedding` model	`openAiLocation`, `chatModelName`, `chatModelVersion`, `chatDeploymentSku`, `chatCapacity`, `embeddingModelName`, `embeddingModelVersion`, `embeddingDeploymentSku`, `embeddingCapacity`
Search	Azure AI Search with an index, a data source, a processing pipeline (skillset) and an indexer	`searchSku`, `searchReplicaCount`, `searchPartitionCount`, `semanticSearch`
Documents	Storage account with a `documents` container	`storageSku`
Secrets	Key Vault	`keyVaultSku`
Monitoring	Log Analytics and Application Insights	`logAnalyticsRetentionDays`, `logAnalyticsDailyQuotaGb`

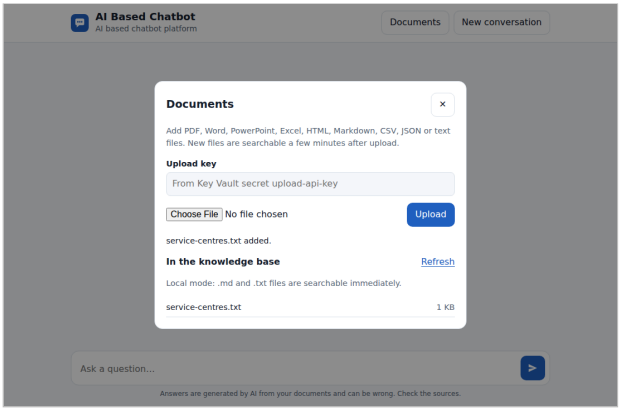
More detail: Architecture document

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Screens



Chat with sources



Adding documents

The screenshots were taken in local mode (see "Trying it without Azure"), where answers are extracts from sample documents. In a real deployment, the answers are written by the language model.

Trying it without Azure

Local mode runs the chat page on your own computer without any Azure services. It searches a few sample documents by keyword, and instead of a language model it quotes the best matching sentences. It is meant for trying the page out and for development:

```
make setup     # creates a Python environment and installs everything
make run       # opens the chatbot at http://localhost:8000
make test      # runs the automated tests
```

More detail: Local development guide

Prerequisites

Requirement	Notes
Azure subscription	You need the **Owner** role on the resource group, or **Contributor** plus **User Access Administrator**, because the template gives the services access to each other.
Azure OpenAI access and quota	Enough quota for the chat and embedding models, with the chosen deployment types, in the Azure OpenAI region. Check it in *Azure AI Foundry → Quotas*. The `dev` preset needs 30K TPM for each model.
Resource providers	These must be registered in the subscription: `Microsoft.Web`, `Microsoft.CognitiveServices`, `Microsoft.Search`, `Microsoft.Storage`, `Microsoft.KeyVault`, `Microsoft.OperationalInsights` and `Microsoft.Insights`. Register a missing one with `az provider register -n <name>`.
Tools (script deployment only)	Azure CLI 2.60 or later, Python 3.10 or later, and Bash or PowerShell 7. The repository's development container already has all of them.

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Choosing SKUs

All service sizes are kept in settings files in the `Deployment/ARM/parameters/` folder, not in the templates. There are three presets:

Preset	Meant for	Main choices
`dev`	Trials and development	App Service B1; AI Search Basic with 1 replica; free semantic ranking; 30K TPM for each model; locally redundant storage (LRS)
`standard`	A small production workload	App Service P0v3; AI Search Basic with 2 replicas (uptime guarantee for searches); standard semantic ranking; 50K TPM for each model; zone-redundant storage (ZRS)
`prod`	More documents and more users	App Service P1v3 with 2 instances; AI Search Standard (S1) with 2 replicas; the larger `text-embedding-3-large` model; 150K TPM for chat and 120K TPM for embeddings; chat processed within the Data Zone; logs kept for 90 days

What each setting controls:

Setting	What it affects	Allowed values
`location`	The region for every service except Azure OpenAI	Any region that offers the services
`appServicePlanSku`, `appServiceInstanceCount`	Web app cost and capacity, and its uptime guarantee	`B1`-`B3`, `S1`-`S3`, `P0v3`-`P3v3`; 1 to 10 instances
`openAiLocation`	Where questions and documents are processed by Azure OpenAI, and which models are available	A region that offers both models
`openAiSku`	Azure OpenAI pricing tier	`S0`
`chatModelName`, `chatModelVersion`	Answer quality, speed and price	Any chat model offered in the Azure OpenAI region
`chatDeploymentSku`, `embeddingDeploymentSku`	Where data is processed, and how much quota is available by default	`GlobalStandard` (any Azure region), `DataZoneStandard` (within the US or EU data zone), `Standard` (within the chosen region)
`chatCapacity`, `embeddingCapacity`	The most text processed per minute, in thousands of tokens. Must fit your quota.	1 to 1000
`embeddingModelName`	Search quality, index size and cost. Changing it later means rebuilding the index.	`text-embedding-3-small`, `text-embedding-3-large`
`searchSku`	Index storage, limits and price	`basic`, `standard`, `standard2`, `standard3`

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Setting	What it affects	Allowed values
`searchReplicaCount`	How many questions can be searched at once; 2 or more for an uptime guarantee on searches	Basic: 1 to 3; Standard tiers: 1 to 12
`searchPartitionCount`	How much document content the index can hold	Basic: 1 to 3; Standard tiers: 1, 2, 3, 4, 6 or 12. Replicas × partitions must be 36 or less.
`semanticSearch`	Extra ranking of search results by meaning, and its cost	`disabled`, `free` (1,000 searches a month), `standard` (paid beyond the free 1,000)
`storageSku`	How many copies of the documents Azure keeps, and where	`Standard_LRS`, `Standard_ZRS`, `Standard_GRS`, `Standard_RAGRS`, `Standard_GZRS`, `Standard_RAGZRS`
`keyVaultSku`	Key Vault tier	`standard`, `premium`
`logAnalyticsRetentionDays`, `logAnalyticsDailyQuotaGb`	How long logs are kept, and the most log data accepted per day	30 to 730 days; GB per day, or `-1` for no limit

How to change the sizes:

1. Edit a preset file, or copy one under a new name. For example, copy `dev.parameters.json` to `uat.parameters.json` and deploy with `-e uat`.
2. Or change single values when you deploy, without editing any file: `deploy.sh -e dev --set searchSku=standard --set chatCapacity=60`, or `deploy.ps1 -Environment dev -Set searchSku=standard,chatCapacity=60`.
3. Check your choices: `python3 Deployment/scripts/validate_skus.py -e dev --check-region`. The deployment scripts always run this check first. It rejects values the template does not allow and search sizes Azure does not accept. With `--check-region` (after `az login`), it also checks that your regions offer the services, models and deployment types, and warns when your Azure OpenAI quota looks too low. Example of the messages it shows:

```
The SKU selection is not valid:
  - searchReplicaCount=5: the basic tier allows 1-3 replicas
  - chat model gpt-4.1-mini version 2025-04-14 with deployment type Standard is not offered
in 'centralindia'.
    List what is: az cognitiveservices model list -l centralindia -o table
```

The Deploy to Azure button offers the same choices as drop-down lists and sliders in the portal. The full list of settings, with the value in each preset, is in the SKU reference.

Deployment options

1. Deploy to Azure button

Click the button near the top of this document. The Azure portal opens a form with these tabs:

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Tab	What you choose
Basics	Subscription, resource group, region, a short name prefix (3 to 8 lowercase letters or digits) and the environment name
Azure OpenAI	The Azure OpenAI region; for the chat model and the embedding model: name, version, deployment type and capacity
Azure AI Search	Tier; replicas and partitions (only the values the tier allows); semantic ranking
Hosting and data	App Service plan and number of instances; storage redundancy; Key Vault tier; log retention and daily log limit
Application	Whether to install the application code (default **Yes**: App Service downloads and builds the `main` branch of the repository); the repository address and branch; whether document upload is allowed

It creates: Log Analytics, Application Insights, a storage account with a `documents` container, a Key Vault with a newly generated upload key, Azure OpenAI with the two models, Azure AI Search, an App Service plan and web app, the access permissions between them, and (unless turned off) the application. Then follow the steps in "After deployment".

2. Script (Bash)

```
az login
az account set --subscription "<subscription>"
Deployment/scripts/deploy.sh -g rg-chatbot-dev -e dev --seed
```

3. Script (PowerShell)

```
az login
./Deployment/scripts/deploy.ps1 -ResourceGroup rg-chatbot-dev -Environment dev -Seed
```

Both scripts check your size choices, create the resource group, deploy the template, give you access to add documents, install the application, and wait until it reports that it is healthy. With `--seed` (`-Seed` in PowerShell) they also load the sample documents.

Option (Bash / PowerShell)	What it does
`--set` / `-Set`	Change single size settings for this deployment
`--skip-infra` / `-SkipInfra`	Install only the application code
`--skip-code` / `-SkipCode`	Create only the Azure services
`--skip-region-check` / `-SkipRegionCheck`	Skip the check against your region (the basic checks still run)
`-l` / `-Location`	Region for the resource group
`-s` / `-Subscription`	Azure subscription to use

You can safely run a script again: it updates what is already there and keeps the existing upload key. (Deploying again with the button creates a new upload key.)

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4. GitHub Actions

The repository includes a workflow that deploys a chosen preset from GitHub, without storing any passwords. It needs a one-time setup, described in the Deployment guide.

After deployment: checking it works and adding documents

1. **Check that it is running:** open `https://<web app address>/healthz`. You should see `"status": "ok"` and `"searchSetup": "ready"`. On its first start, the app creates the search index and indexer. If it shows `"pending"`, wait a few minutes. If it shows `"failed"`, the access permissions were probably not active yet: wait a few minutes and restart the web app.
2. **Get the upload key:**

   ```
   az keyvault secret show --vault-name <key-vault> -n upload-api-key --query value -o tsv
   ```

   The scripts give the person who deploys access to read it. After a deployment with the button, first give yourself the *Key Vault Secrets User* role on the Key Vault.
3. **Add documents**, in one of these ways:
   - **On the chat page:** select **Documents**, paste the upload key, choose a file and select **Upload**. The window shows the indexing status and any errors for individual files.
   - **With a script** (for many files or whole folders): `Deployment/scripts/ingest.sh -g <resource-group> ./my-docs ./faq.pdf`, or `./Deployment/scripts/ingest.ps1 -ResourceGroup <resource-group> -Path ./my-docs`.
   - **With any tool that copies files to Azure Storage** (for example Azure Storage Explorer or AzCopy), into the `documents` container. The indexer picks them up within 30 minutes.
4. **Wait a few minutes**, then ask a question that your documents answer.
5. If you loaded the sample documents (`--seed`), remove them: delete them from the `documents` container (for example in the Azure portal or Storage Explorer), or call `DELETE /api/documents/<name>` with the upload key. The indexer removes deleted files from the index on its next run. The chat page cannot delete documents.

Full steps and troubleshooting: Deployment guide

Using the chatbot

- Type a question and press **Enter** (Shift+Enter starts a new line). The page shows *Searching the documents…* while it works.
- Answers show source numbers. Select a number, or a source under the answer, to see the passage it came from.
- Ask follow-up questions naturally, for example "and for washing machines?". The last six questions and answers are taken into account.
- If your documents do not cover a question, the chatbot says so instead of guessing.
- **New conversation** clears the conversation. The conversation is kept only in the current browser tab.

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- If something goes wrong (for example the service is busy or the connection drops), your question stays in the box and **Try again** sends it again.
- Developers can find the technical description of the programming interface at `/api/docs`.

Cost, security and clean-up

Cost notes

- **Charged all the time, even with no users:** the App Service plan (per instance) and Azure AI Search (per replica and partition). These are most of the fixed monthly cost. The `dev` preset uses the smallest practical sizes.
- **Charged only when used:** Azure OpenAI (by the amount of text in questions, answers and indexed documents), semantic ranking beyond the free 1,000 searches a month on the `standard` plan, document storage (by size), and log data (limited by `logAnalyticsDailyQuotaGb`).
- The `text-embedding-3-large` model costs several times more than `text-embedding-3-small` and needs about twice as much index space for its vectors.
- Check current prices for your region before deploying: https://azure.microsoft.com/pricing/calculator/

Security notes

- **No keys between services:** the services use managed identities, each with only the access it needs, and key-based access is switched off on Azure OpenAI, Azure AI Search and Storage. The upload key is the only secret, and it is kept in Key Vault.
- **Uploads are protected** by the upload key and by checks on the file type and size (16 MB at most), because whatever is uploaded becomes source material for every user's answers.
- **The chat page has no sign-in by default.** Anyone with the address can ask questions (each internet address is limited to about 20 questions a minute). To limit it to your organisation, turn on App Service Authentication with Microsoft Entra ID: *Web app → Authentication → Add identity provider → Microsoft*, then require sign-in. The Deployment guide has the matching command.
- The services can be reached over the internet but require Microsoft Entra ID sign-in. Private network connections and a web application firewall are not part of this template.
- More detail: SECURITY.md

Clean-up

```
az group delete --name rg-chatbot-dev --yes --no-wait
```

Azure keeps deleted Key Vaults and Azure OpenAI accounts for a while (soft delete). To deploy again with the same names (same resource group and prefix), remove them permanently first:

```
az keyvault purge --name <key-vault-name>
az cognitiveservices account purge --name <openai-account-name> --resource-group rg-chatbot-dev --location <openai-location>
```

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Troubleshooting

What you see	What to do
`The SKU selection is not valid`	Read the values it lists, then change the preset or use `--set`. If a model is "not offered" in a region, see what is available with `az cognitiveservices model list -l <region> -o table`, or choose another `openAiLocation`.
`InsufficientQuota` during deployment	Lower `chatCapacity` or `embeddingCapacity`, choose another deployment type or region, or ask Azure for more quota.
`AuthorizationFailed` when permissions are created	You need Owner, or Contributor plus User Access Administrator, on the resource group.
A name already exists, or a Key Vault or Azure OpenAI account was deleted earlier	Remove it permanently (see "Clean-up").
`/healthz` shows `"searchSetup": "failed"`	Usually the access permissions were not active yet. Wait a few minutes and restart the web app. `searchSetupDetail` gives the reason.
Upload says the key is not available yet	The web app could not read the key from Key Vault yet. Wait a few minutes and restart the web app.
"The service could not answer right now"	Look in Application Insights → Failures for the cause (for example a wrong model name, missing permission or a retired model).
"Too many questions in a short time"	The limit of about 20 questions a minute from one internet address was reached. If many users share one internet address, raise `RATE_LIMIT_PER_MINUTE`.
Answers never find a document	Check the indexing status and errors in the Documents window. Scanned PDFs without selectable text cannot be read (text recognition is not included).
The button deployment fails while installing the code from GitHub	The Azure services are already in place. Install the code with `deploy.sh -g <resource-group> -e dev --skip-infra`.

More: Deployment guide → Troubleshooting

Repository layout

Folder	Contents
`Code/backend`	The web app: questions and answers, search, sources, document upload, and its automated tests
`Code/frontend`	The chat page
`Deployment/bicep`	The infrastructure definition, one file per group of services
`Deployment/ARM`	The template used by the button and the scripts, the presets (`parameters/`), and the portal form
`Deployment/scripts`	Deployment, document loading and size-checking scripts for Bash and PowerShell

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Folder	Contents
`Data/sample_docs`	Made-up sample documents for local mode and first tests
`Documents`	Architecture, deployment guide, settings and local development guides
`imgs`	Architecture diagram and screenshots
`.github/workflows`	Automatic checks on every change, and the manual deployment workflow
`.devcontainer`	A ready-made development environment with all the tools installed

Current status

Item	Status
Answers with sources and follow-up questions; document upload and loading by script; the template, button, scripts and presets	Built; tested on a developer machine and with simulated Azure services
First deployment to a live Azure subscription (including model availability in the default regions and installing the code from the button)	Still to do: needs a subscription with Azure OpenAI quota
Sign-in built into the template, private network connections, text recognition for scanned PDFs	Planned or available on request

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